Vapour-Liquid Equilibrium of Mixtures

Module 4 · 2105603 Advanced Chemical Engineering Thermodynamics

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A parameter set is not a result.

A result is a parameter set plus a statement of where it may be used.

What this module establishes

Four descriptions of the same equilibrium, the behaviour they produce, then the full working cycle: data, consistency, regression, a stated domain of validity — and the route back into the equation of state.

The chain of reasoning

An equation of state gives P-V-T behaviour → equilibrium is governed by chemical potential → fugacity gives chemical potential a usable form → equality of fugacity predicts phase equilibrium.

Module 4 is where that chain meets measurement — and, in Section 4.7, where it returns to the equation of state it started from.

Seven questions

- Which description of VLE, and what does it cost?
- What kind of mixture is this, and will it azeotrope?
- What is actually measured, and how well?
- Can this dataset be believed?
- Which model, fitted against what?
- Where may the answer be used?
- When does the activity route run out?

PART I

Four descriptions of one equilibrium

Section 4.1 Raoult, modified Raoult, gamma-phi, phi-phi

Each rung adds physics the rung below omitted, and each addition costs something.

Raoult's law, and its two assumptions

State the two assumptions separately, because they fail separately.

$$y_i P = x_i P_i^{\mathrm{sat}}$$

Assumption 1 — ideal liquid

Every molecule sees the same environment whatever the composition. True only when the two species are chemically almost identical: benzene and toluene, hexane and heptane.

Assumption 2 — ideal vapour

$\hat{\varphi}_i = 1$. Fails at high pressure, and also at low pressure when the vapour associates. Acetic acid dimerises in the vapour at one atmosphere.

What it costs

Nothing but $P_i^{\mathrm{sat}}(T)$. No binary parameter, no data, no fitting. That is why it survives as the first estimate.

Modified Raoult's law

Liquid non-ideality admitted; the vapour still treated as an ideal gas.

$$y_i P = x_i \gamma_i P_i^{\mathrm{sat}}$$

Physical meaning

γ_i measures how much the liquid environment differs from the pure liquid. It is a property of the *mixture*, derived from an excess Gibbs energy, and it carries the reference state with it — Lewis-Randall here, so $\gamma_i \to 1$ as $x_i \to 1$.

Engineering implication

This is the description behind almost every distillation column ever designed at moderate pressure. Two fitted parameters per binary pair, from binary VLE data. Everything in Sections 4.3 to 4.5 is about where those two numbers come from and what they are worth.

The gamma-phi formulation

Both phases non-ideal — but described by two different kinds of model.

$$y_i \hat{\varphi}_i P = x_i \gamma_i P_i^{\text{sat}} \varphi_i^{\text{sat}} \exp\left[\frac{v_i^L \left(P - P_i^{\text{sat}}\right)}{RT}\right]$$

$\hat{\varphi}_i$

Vapour fugacity coefficient, from an equation of state. The two-term virial equation is enough below a few bar.

φ_i^{sat}

The same quantity for the pure saturated vapour. It is the reference the liquid fugacity is measured against, and it does not cancel.

Poynting factor

Corrects the liquid from P_i^{sat} to P. Small below a few bar, but it is not zero, and omitting it silently is not the same as knowing it is small.

The phi-phi formulation

One equation of state for both phases. The only option above the critical temperature of a component.

$$y_i \hat{\varphi}_i^V = x_i \hat{\varphi}_i^L$$

Why it is needed

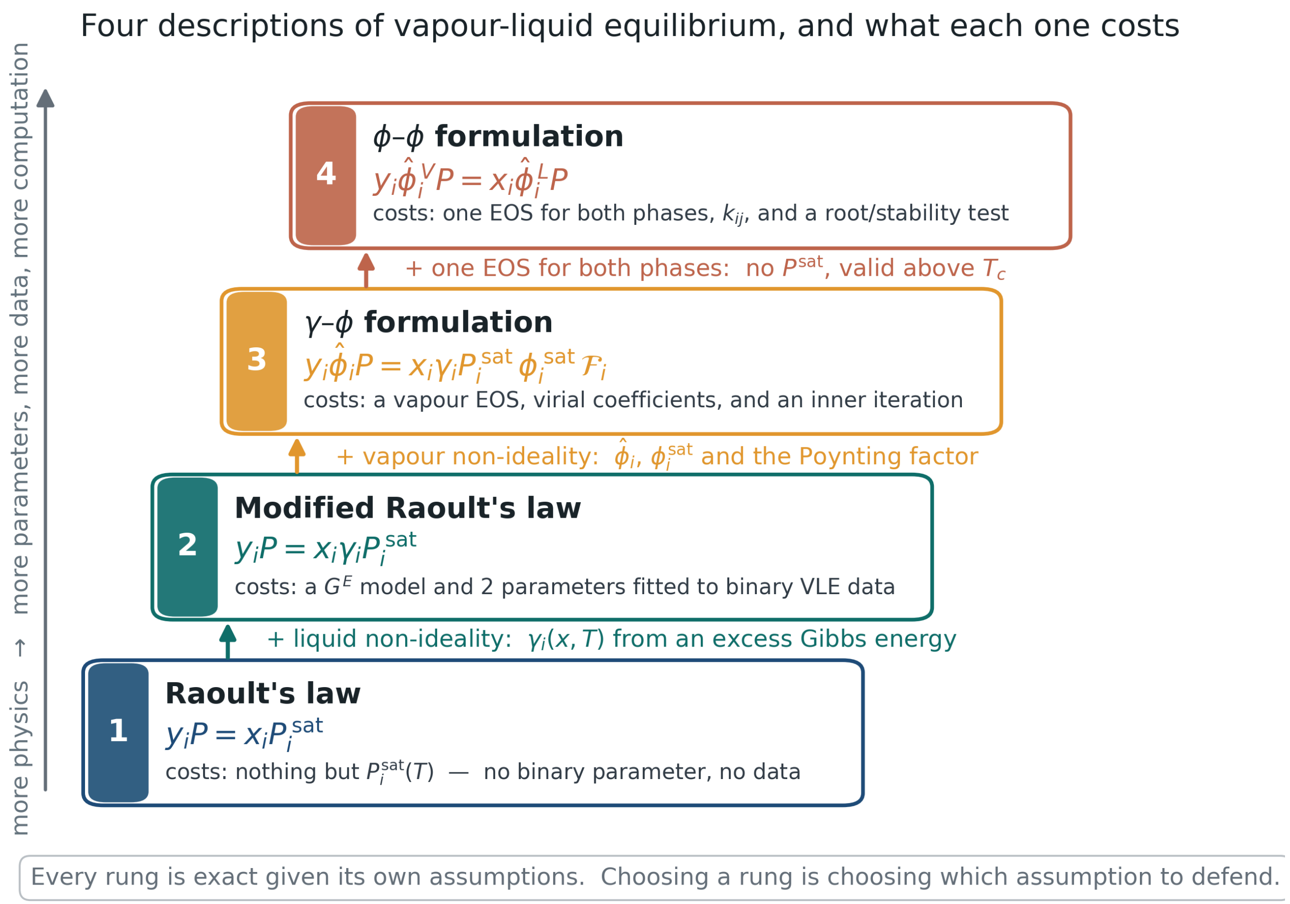
A supercritical component has no P_i^{sat}, so the whole right-hand side of the gamma-phi equation is undefined. Natural gas, hydrogen, supercritical CO_2: there is no vapour pressure to refer to.

What it costs

One equation of state that must describe a dense liquid and a dilute vapour with the same parameters, a binary interaction parameter k_{ij} fitted to data, and a root selection problem — the cubic has three roots and choosing wrongly gives a confidently wrong answer.

The ladder

Each rung is exact given its own assumptions. Choosing a rung is choosing which assumption to defend.



F4.1 · The four descriptions, with the physics added and the cost incurred at each step.

Choosing a description

A decision table, not a ranking.

SYSTEM	DESCRIPTION	BECAUSE
Benzene / toluene, 1 atm	Raoult	Chemically similar; no binary data needed
Ethanol / water, 1 atm	Modified Raoult	Strongly non-ideal liquid, near-ideal vapour
Acetic acid / water, 1 atm	Gamma-phi	Vapour associates; $\hat{\varphi} \neq 1$ at 1 atm
Methane / propane, 80 bar	Phi-phi	One component is supercritical
CO_2 / ethanol, 100 bar	Phi-phi with a G^E mixing rule	Supercritical *and* polar — Section 4.7

The last row is the case neither classical description handles. Section 4.7 builds the description that does, and reports honestly what it is worth.

PART II

Solution types and azeotropes

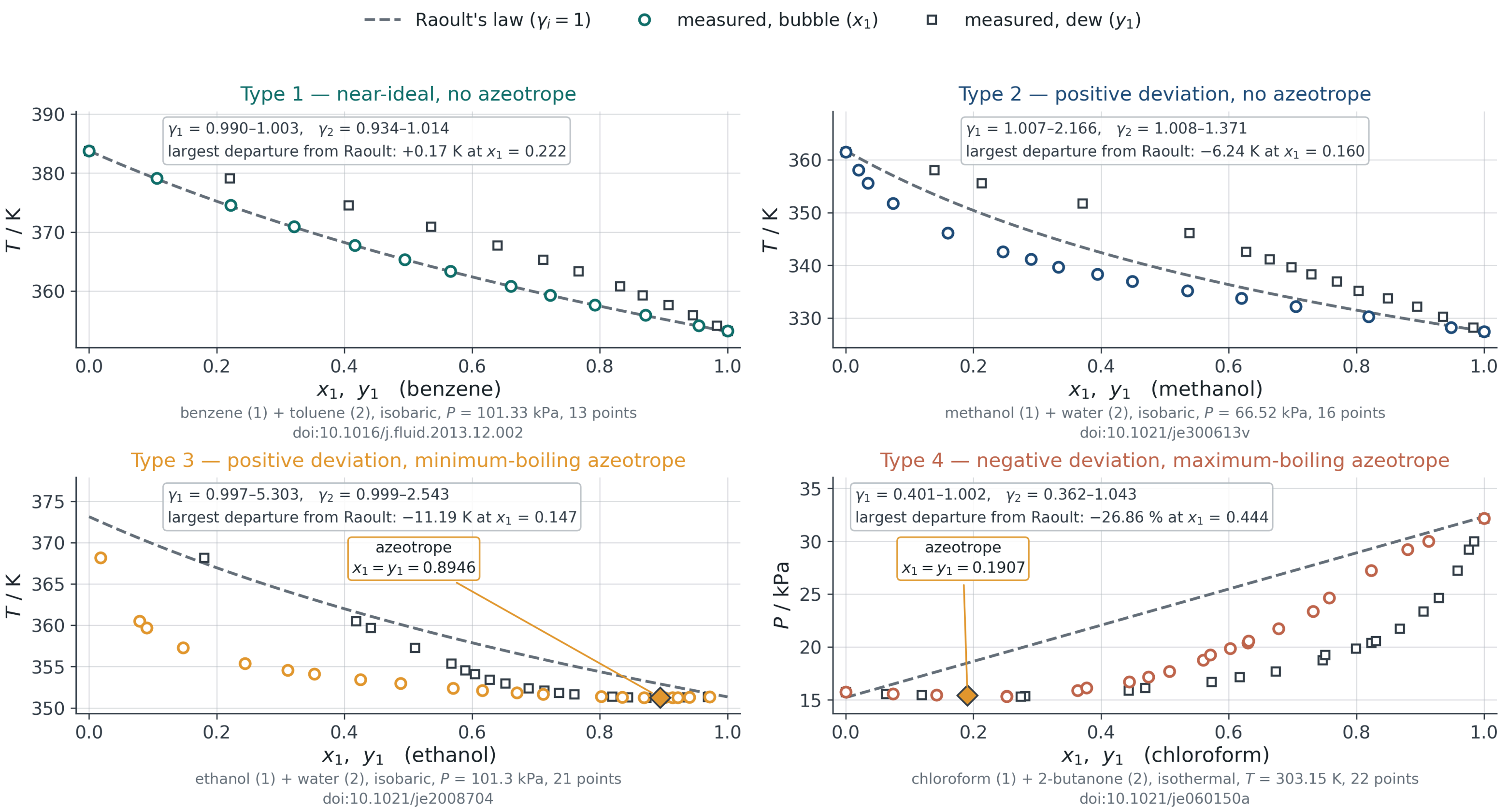
Section 4.2 What the four descriptions produce.

Four classes of behaviour, one criterion, and an azeotrope that belongs to the pressure rather than to the mixture.

Four types, on four measured systems

The classification is one statement about the sign and size of G^E, read against a common Raoult reference.

The four solution types, on four published datasets — the dashed line is Raoult's law with the same Antoine constants throughout



The criterion, derived

An azeotrope is not a category a mixture belongs to. It is a root of one equation, and the root either lies in $(0, 1)$ or it does not.

$$y_1 = x_1 \iff \gamma_1 P_1^{\mathrm{sat}} = \gamma_2 P_2^{\mathrm{sat}} \iff \alpha_{12} \equiv \frac{\gamma_1 P_1^{\mathrm{sat}}}{\gamma_2 P_2^{\mathrm{sat}}} = 1$$

$$\alpha_{12}\Big|_{x_1 \to 0} = \gamma_1^{\infty} \frac{P_1^{\mathrm{sat}}}{P_2^{\mathrm{sat}}}, \qquad \alpha_{12}\Big|_{x_1 \to 1} = \frac{1}{\gamma_2^{\infty}} \frac{P_1^{\mathrm{sat}}}{P_2^{\mathrm{sat}}}$$

The test at the ends

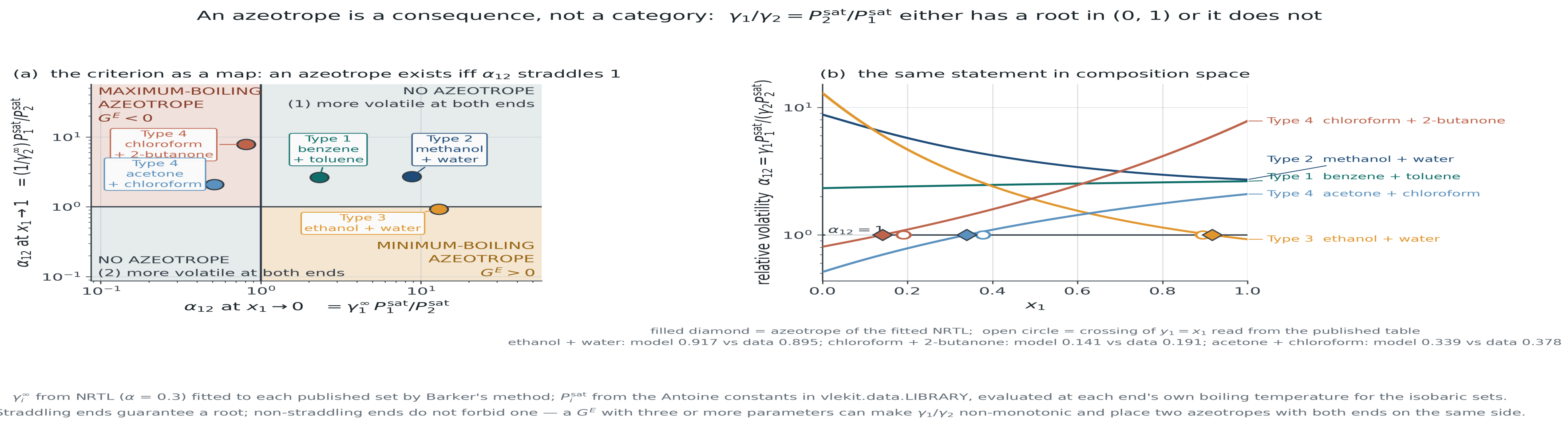
If α_{12} is above 1 at one end and below 1 at the other, a continuous $\alpha_{12}(x_1)$ must cross, and the crossing is the azeotrope. Only the two γ^{∞} and the vapour-pressure ratio are needed — no interior data at all.

Two components, one verdict

Methanol + water has a genuine positive deviation, $\gamma_1^{\infty} = 2.41$, and no azeotrope: $P_1^{\mathrm{sat}}/P_2^{\mathrm{sat}}$ is already 3.6 to 4.3, so α_{12} never approaches 1. **Type 2 and Type 3 differ by the vapour pressures, not by the deviation.**

The criterion as a map

Two end-point volatilities place a system in a quadrant. Five published systems, five correct verdicts.



What was predicted

γ^∞ from an NRTL fitted by Barker to each set, and P^{sat} from Antoine. Ethanol + water: $\alpha = 12.96$ and 0.92 — straddles, so minimum-boiling. Chloroform + 2-butanone: 0.81 and 7.83 — straddles from below, so maximum-boiling.

What checked it

The measured relative volatility, from x_1 and y_1 alone, with no P^{sat} and no model: it crosses 1 for exactly the three systems the map predicted. **Five out of five**, by two routes sharing no numbers.

F4.12 · Five published systems on the criterion map, and the same statement in composition space. The two routes are independent.

Minimum-boiling and maximum-boiling

Gibbs-Konovalov: $\mathrm{d}P/\mathrm{d}x_1 = 0$ exactly where $y_1 = x_1$. An azeotrope is always a stationary point; the sign of G^E decides which kind.

The argument, without a model

$\gamma_i > 1$ puts $P = x_1\gamma_1 P_1^{\mathrm{sat}} + x_2\gamma_2 P_2^{\mathrm{sat}}$ **above** the Raoult straight line at every interior composition. A continuous curve that starts and ends on a line and lies above it in between has an interior maximum.

Maximum in P is minimum in T

Higher pressure at fixed T means the mixture boils below both pure components. So positive deviation gives a **minimum-boiling** azeotrope, and negative deviation reverses every step of the argument to give a **maximum-boiling** one.

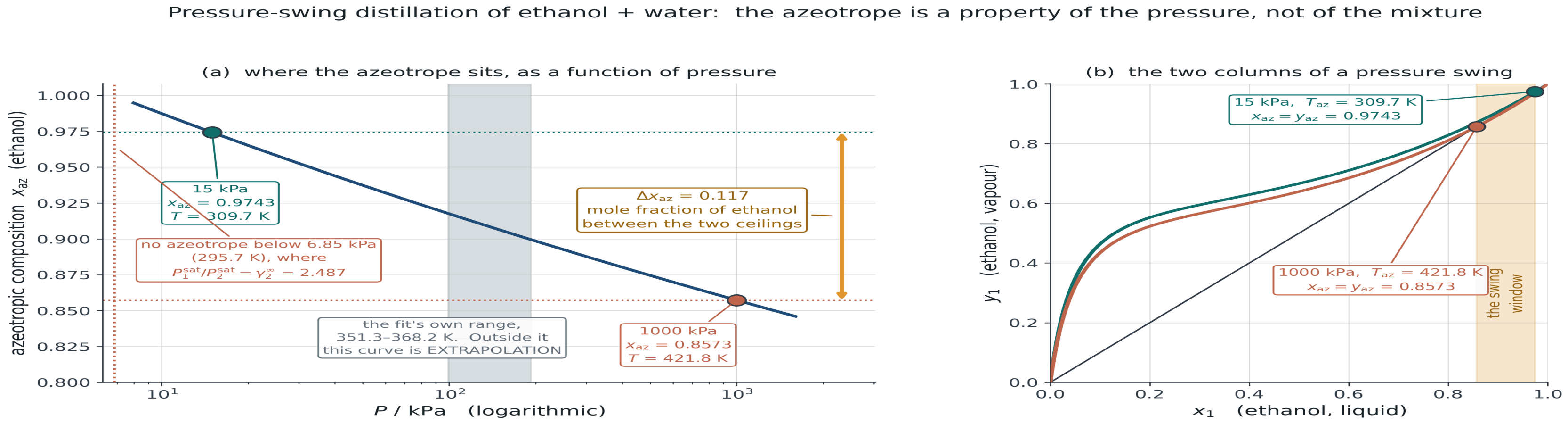
Confirmed, not quoted

$\mathrm{d}P/\mathrm{d}x_1$ at the computed azeotrope is -2×10^{-9} kPa for ethanol + water, $+2 \times 10^{-9}$ for chloroform + 2-butanone and -1×10^{-9} for acetone + chloroform. Konovalov to nine decimals.

And the uncomfortable corollary: the P-x curve is ***flat*** *at the azeotrope, so the pressure carries almost no information about where it sits. The same fits that reproduce P to 0.32 kPa misplace the azeotrope by $+0.023$, -0.050 and -0.038 in x_1. If the azeotropic composition is the number your design needs, it must be in the objective function.*

The azeotrope moves with pressure

γ_1/γ_2 depends on composition, $P_2^{\text{sat}}/P_1^{\text{sat}}$ only on temperature. Move the pressure and the balance point moves.



MODEL: NRTL (α = 0.3, τ_{12} = -0.1712, τ_{21} = +1.9330) fitted by Barker's method to Kamihama et al. 2012, doi:10.1021/je2008704 — 21 points, 101.3 kPa, 351.3–368.2 K, rms δT = 0.059 K. P_i^{sat} from vlekit.data.LIBRARY.
THE PARAMETERS ARE HELD FIXED AS P IS VARIED. τ_{ij} carries no temperature dependence of its own, so in this calculation the whole shift comes from $P_1^{sat}/P_2^{sat}(T)$; the real G^E also changes with T, and a fit at one pressure cannot know how.
For scale: at 101.3 kPa this model puts the azeotrope at x_1 = 0.9172 where the published table gives 0.8946, a difference of 0.0226 in mole fraction.

The window

Ethanol + water: $x_{\text{az}} = 0.9743$ at 15 kPa and 0.8573 at 1000 kPa — a window of **0.117**. Below 6.85 kPa (295.7 K) there is no azeotrope at all, because $P_1^{\text{sat}}/P_2^{\text{sat}}$ has fallen to $\gamma_2^\infty = 2.487$.

What is assumed

NRTL was fitted at 101.3 kPa over 351.3–368.2 K and its τ_{ij} carry no T dependence, so the whole shift comes from $P^{\text{sat}}(T)$. The 1000 kPa point sits 53.6 K outside that range: **an extrapolation.**

F4.13 · NRTL by Barker on doi:10.1021/je2008704, held fixed as P varies.

When a pressure swing is worth building

Two columns at two pressures beat the azeotrope only if the azeotrope moves. Two Type 3/4 systems, two answers.

Ethanol + water: it moves

x_{az} runs 0.9743 at 15 kPa to 0.9172 at 101 kPa — 0.057 over the range a real vacuum column would use, and 0.117 out to 1000 kPa. The prize is real but modest, which is why industry usually prefers extractive or azeotropic distillation to a second column plus a vacuum system.

Acetone + chloroform: it does not

x_{az} runs 0.3402 at 90 kPa to 0.3469 at 25 kPa — **0.007 over a factor of 3.5 in pressure**. A pressure swing on this pair is not a separation, it is a rounding error.

The reason is in the criterion. Acetone + chloroform has $P_1^{\text{sat}}/P_2^{\text{sat}} \approx 1.18$ at both ends, barely moving with temperature, while γ_1/γ_2 is steep in composition, so the root hardly shifts. Ethanol + water has two components of similar volatility and a γ_1/γ_2 that is nearly flat near $x_1 = 0.9$: a small change in the vapour-pressure ratio then moves the crossing a long way.

PART III

From measurement to activity coefficient

Section 4.3   A reported γ is not a measurement.

It is four measured numbers and two assumptions.

What a VLE apparatus measures

Four numbers per point, each with its own uncertainty, and none of them is γ.

Temperature

Calibrated thermometer in the boiling liquid. Typically ± 0.05 K. Its error enters through $P_i^{\mathrm{sat}}(T)$, which is exponential in T.

Pressure

Manometer or transducer. Typically ± 0.05 kPa. Usually the *best* measured variable, which is why Barker's method uses it.

Liquid composition

Sample and analyse — refractometry, gas chromatography, density. Typically ± 0.001 mole fraction, and only if the calibration is honest.

Vapour composition

Condense and analyse. The hardest measurement: partial condensation, holdup, and entrainment all bias it. Typically ± 0.002, often worse.

Computing gamma

Two assumptions convert four measurements into an activity coefficient.

$$\gamma_i = \frac{y_i \Phi_i P}{x_i P_i^{\mathrm{sat}}(T)}, \qquad \Phi_i = \frac{\hat{\varphi}_i}{\varphi_i^{\mathrm{sat}}} \exp\left[-\frac{v_i^L (P - P_i^{\mathrm{sat}})}{RT}\right]$$

Assumption A — the vapour

Setting $\Phi_i = 1$ is modified Raoult. For acetone/chloroform at 50 °C the virial correction shifts $\ln \gamma_1$ by up to 0.008 — small, but **systematic**, and a consistency test reacts to systematic errors, not to scatter.

Assumption B — the vapour pressure

P_i^{sat} comes from a correlation, not from this experiment. Antoine coefficients are copied between references with unit errors more often than any other quantity in this subject. Check the correlation against the normal boiling point before you trust one activity coefficient.

Taking γ out of a published table

The paper gives you four columns. Three of the things you need are not in it.

1. **Copy the table with its DOI and its stated uncertainties.**
$u(T)$, $u(P)$, $u(x)$, $u(y)$ are part of the data. A table transcribed without them cannot be reduced honestly.

2. **Check P_i^{sat} against the paper's own end points.**
The pure rows are a free calibration of your Antoine constants — and they are the check almost nobody runs.

3. **Choose Φ_i and say which.**
Ideal vapour, virial $\hat{\varphi}$, or the full three factors. The next two slides say how to decide rather than how to prefer.

4. **Report γ with an uncertainty, then test it.**
The consistency tests of Section 4.4 are the reason the reduction was worth doing.

Step 2, on chloroform + 2-butanone at 303.15 K (doi:10.1021/je060150a): the table's pure end points read 32.19 and 15.74 kPa, and the stored Antoine constants give 32.33 and 15.22 — $+0.44$ *% on chloroform but* -3.28 *% on 2-butanone. That error goes straight into every* γ_2*, and it is larger than every vapour-phase correction on the next slide put together.*

Where the error actually comes from

Propagate, and the ranking is not the one students expect.

$$\sigma^2(\ln\gamma_1) = \left(\frac{\sigma_y}{y_1}\right)^2 + \left(\frac{\sigma_P}{P}\right)^2 + \left(\frac{\sigma_x}{x_1}\right)^2 + \left(\frac{\mathrm{d}\ln P_1^{\mathrm{sat}}}{\mathrm{d}T}\,\sigma_T\right)^2$$

The temperature term

P^{sat} is exponential in T, so $\mathrm{d}\ln P^{\mathrm{sat}}/\mathrm{d}T$ is of order $0.03\ \mathrm{K}^{-1}$ near the normal boiling point. A 0.05 K error is then worth 0.0015 in $\ln\gamma$ — comparable to the composition term.

The dilute ends

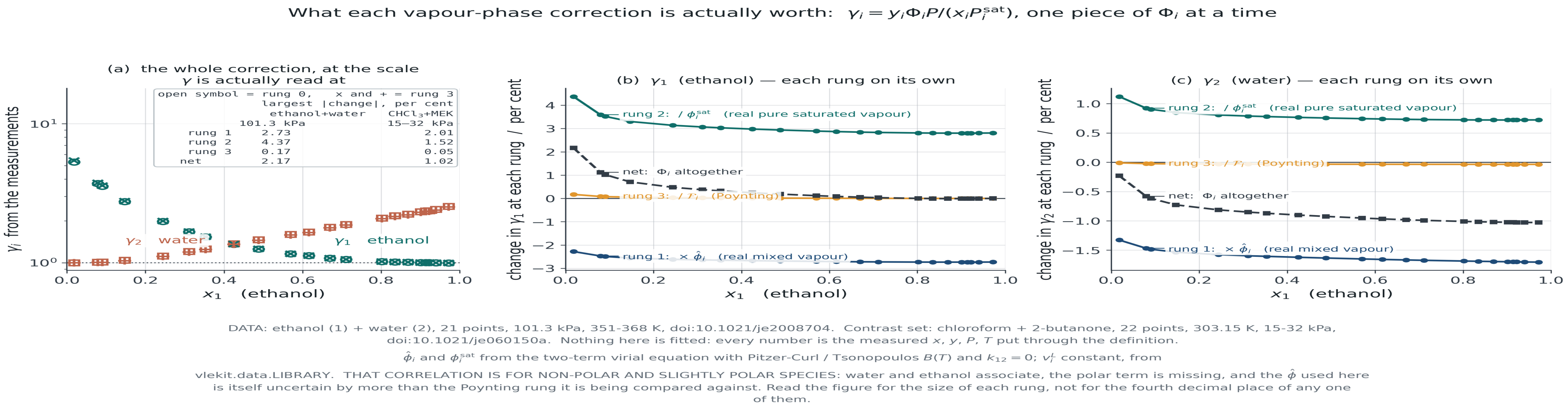
σ_x/x_1 blows up as $x_1 \to 0$. At $x_1 = 0.02$ a 0.001 error in composition is 5 % in γ_1. That is exactly where the model is least constrained and where γ^∞ is read off.

The consequence

Quoting γ to four figures from a thermometer read to 0.1 K is arithmetic, not measurement. Report the uncertainty or do not report the figure.

The corrections, switched on one at a time

Φ_i is three factors, not one. Applied together they largely cancel; applied singly they do not.



The sizes, chloroform + 2-butanone at 15–32 kPa

$\hat{\varphi}_i$ alone moves $\ln\gamma$ by 0.014 and 0.020. Adding φ_i^{sat} moves it **back** by 0.015 and 0.011. Poynting adds 0.0005. Net over all three: 0.009 and 0.010.

Why they cancel

$\hat{\varphi}_i$ in the mixture and φ_i^{sat} for the pure saturated vapour have the same physical origin and, at these pressures, nearly the same value. **Applying one without the other is worse than applying neither** — a correction of one sign with its compensating partner omitted.

F4.15 · Each rung of Φ_i on its own, ethanol + water at 101.3 kPa (doi:10.1021/je2008704) with chloroform + 2-butanone (doi:10.1021/je060150a) as the low-pressure contrast. Nothing here is fitted.

The rule: compare the correction with the noise

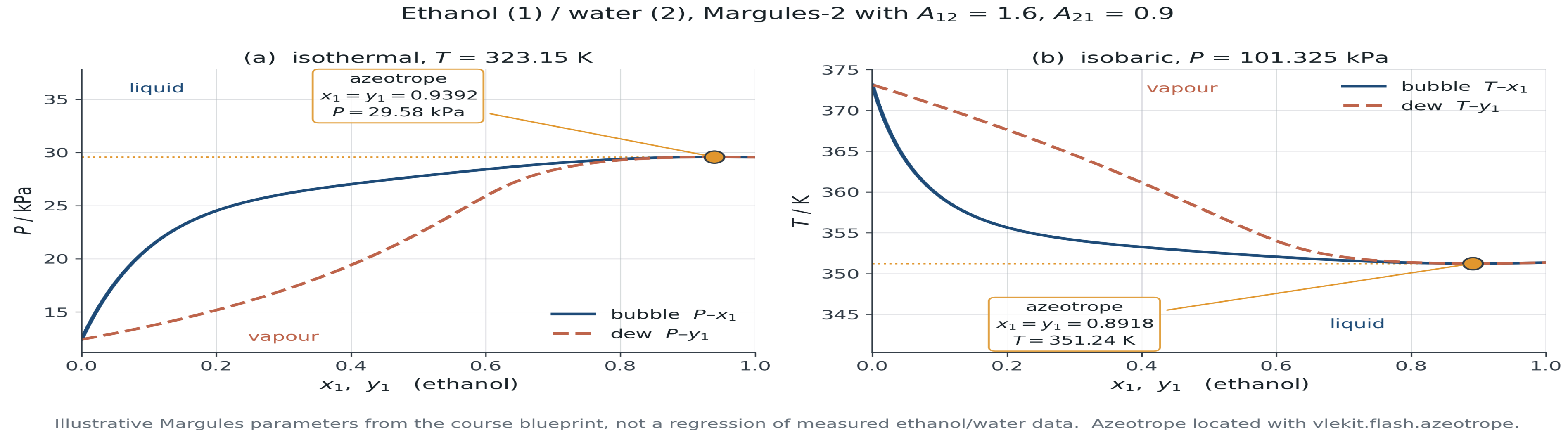
Not “correct above one bar”. Compare the size of the correction with the uncertainty of the measurement it is correcting, on your own dataset.

SYSTEM	P / KPA	NET $\lvert\Delta \ln \Gamma_1\rvert$	$\Sigma(\ln \Gamma_1)$	RATIO
Benzene + toluene	101.3	0.0329	0.0130	2.53
Acetone + chloroform	101.3	0.0108	0.0064	1.70
Methanol + water	66.5	0.0282	0.0185	1.53
Ethanol + water	101.3	0.0214	0.0215	1.00
Chloroform + 2-butanone	20.4	0.0086	0.0443	0.19

At atmospheric pressure the correction is comparable to or larger than the noise and you may not neglect it. At 20 kPa it is a fifth of the noise and you may — ***and you say so, with the number****. σ is propagated from the uncertainties the papers themselves report.*

Isothermal, isobaric, and the azeotrope

The two kinds of dataset, and the one composition where γ comes for free.



Isothermal is easier to use

At constant T the Gibbs-Duhem equation has no enthalpy term. Isobaric data are more common — that is how a column runs — and harder, because T varies across the composition range.

The azeotrope is a free measurement

At $x_i = y_i$ the equilibrium condition gives $\gamma_1/\gamma_2 = P_2^{\text{sat}}/P_1^{\text{sat}}$ directly, with no vapour sample at all. In an old dataset it is often the only point that can be trusted without qualification.

F4.2 · The same system as P-x-y and as T-x-y, with the azeotrope marked on both.

PART IV

Thermodynamic consistency

Section 4.4 Can this dataset be believed?

Gibbs-Duhem makes a binary VLE dataset over-determined, so the data can be tested against thermodynamics with no model assumed correct.

The premise

One of the four measured variables is redundant. The size of the prediction error is the test.

$$x_1 \frac{\mathrm{d}\ln\gamma_1}{\mathrm{d}x_1} + x_2 \frac{\mathrm{d}\ln\gamma_2}{\mathrm{d}x_1} = 0 \qquad (\text{constant } T,\ P)$$

Satisfied by construction

Any model derived from a single G^E satisfies this **identically** — the residual is zero to machine precision. Finding otherwise means the code is wrong, not the fluid.

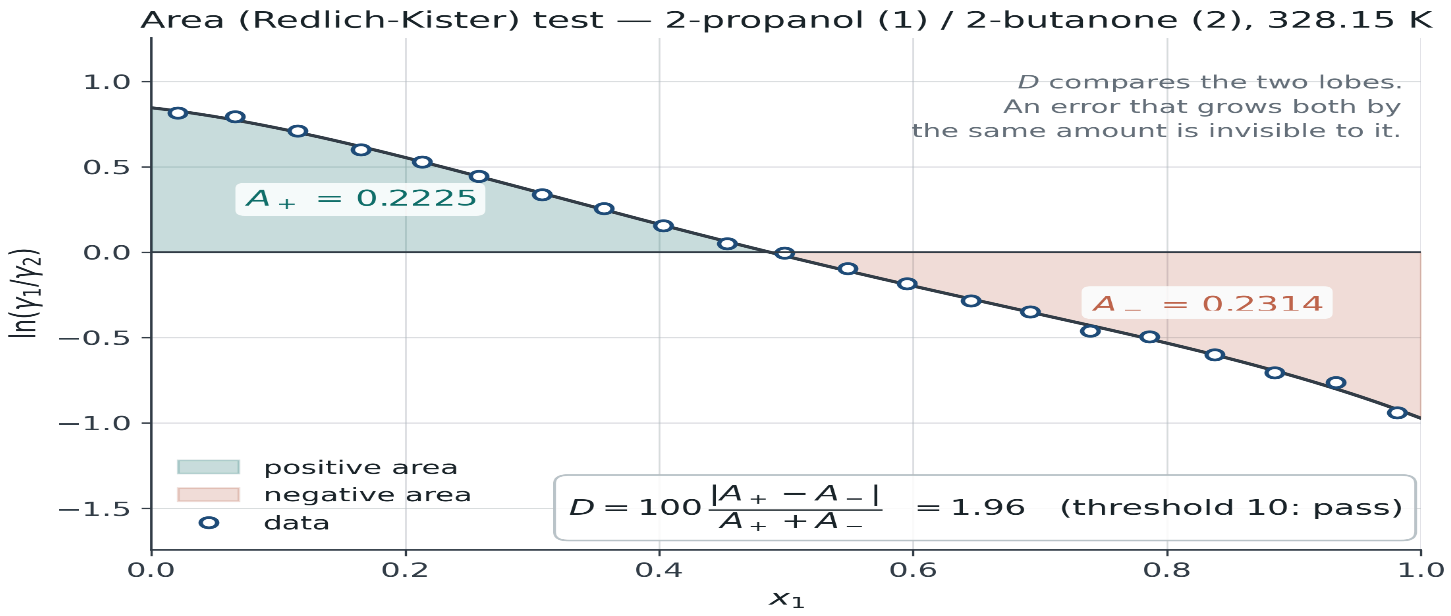
Satisfied by measurement

Data satisfy it only to within their errors. The two statements look alike on a slide and are completely different in practice. Everything that follows is a way of deciding whether the gap is noise or a real error.

The area test

The oldest test, the most quoted, and the weakest.

$$\int_0^1 \ln \frac{\gamma_1}{\gamma_2}\, \mathrm{d}x_1 = 0, \qquad D = 100\,\frac{|A^+ - A^-|}{A^+ + A^-}$$



F4.3 · *The two areas whose difference must vanish, with D reported.*

Why the area test is weak

An integral cannot see an error that changes sign. This is a structural limitation, not a matter of threshold.

The construction

Perturbing y_1 shifts the measured ratio by $\delta \ln(\gamma_1/\gamma_2) = \delta y_1/[\,y_1(1 - y_1)\,]$.

Choose $\delta y_1 = \varepsilon\, y_1(1 - y_1)(2x_1 - 1)$ and the ratio shifts by $\varepsilon(2x_1 - 1)$ — odd about $x_1 = \frac{1}{2}$, so it contributes **exactly nothing** to the integral.

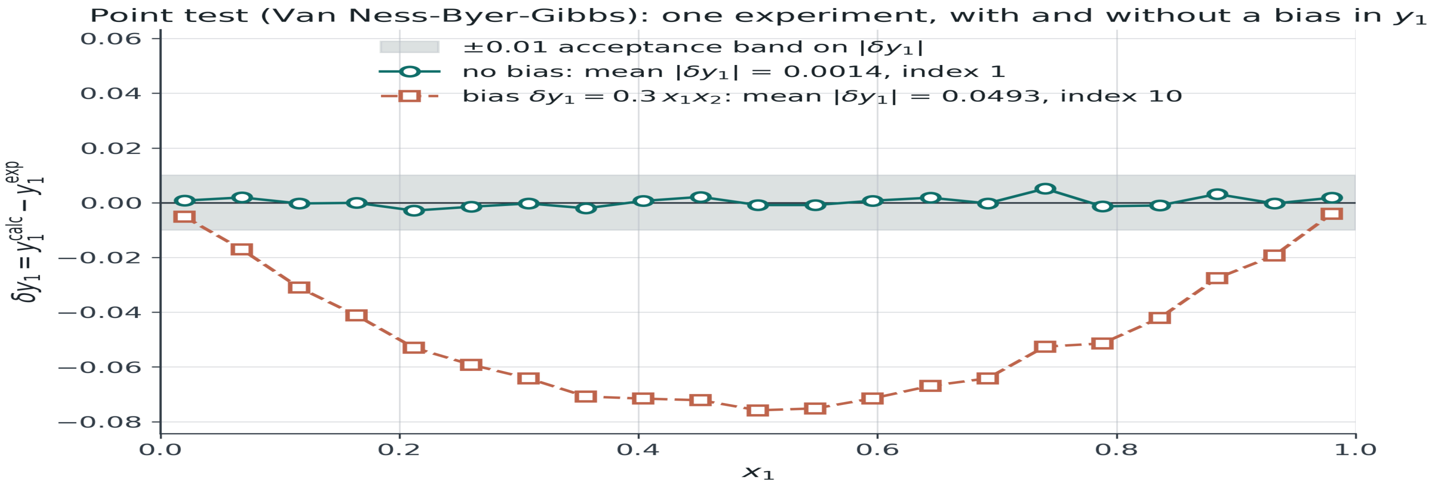
The result

At $\varepsilon = 0.4$ the dataset is grossly wrong. The area test reports $D = 1.3$ against a threshold of 10 — healthier than the merely noisy control — while the point and direct tests both return the worst possible index.

An area test is a necessary condition. It is never a sufficient one.

The point test

Van Ness, Byer and Gibbs: fit P-x only, then predict y. Errors cannot cancel, because nothing is integrated.



2-propanol (1) / 2-butanone (2) at 328.15 K, generated from NRTL(0.35, 0.62); Barker fit to P-x only, y_1 predicted.

The logic

Barker's method uses only P and x. If the data obey Gibbs-Duhem, y is already implied by the P-x curve, so the fit must reproduce it. Mean $|\delta y_1| < 0.01$ is the conventional pass.

The trap

A large residual in P means the **model** failed, not the data. Check $\mathrm{rms}\,\delta P$ before blaming the measurement, and use a model flexible enough to follow the P-x curve.

F4.4 · Point-test residuals for a consistent and an inconsistent dataset on common axes.

The direct test, and the 1-10 index

The same redundancy, expressed in the variable Gibbs-Duhem is actually written in. This is what modern journals expect.

$$\delta = \ln\left(\frac{\gamma_1}{\gamma_2}\right)_{\text{exp}} - \ln\left(\frac{\gamma_1}{\gamma_2}\right)_{\text{fitted}}$$

INDEX	RMS Δ	VERDICT
1 – 3	< 0.075	Good. Publishable without comment
4 – 5	0.075 – 0.125	Marginal. Requires an explanation
6 – 10	> 0.125	Reject, or diagnose the fault

The fitted values come from a Barker fit to P-x alone, so δ measures the data, not the fit — provided the model follows P-x.

Two more tests, and what they add

Different tests are blind to different things. That is the reason to run more than one.

Infinite dilution

Kojima's test compares $(G^E/RT)/x_1x_2$ with $\ln(\gamma_1/\gamma_2)$ at each pure end point, where the two must coincide. Catches the dilute region, which the area test averages away and where γ^∞ is read off.

Herington $D - J$

The area test with an empirical allowance $J = 150\,\Delta T/T_{\text{min}}$ for the temperature variation of an isobaric set. Widely required by referees; Wisniak has shown it both passes bad data and fails good data.

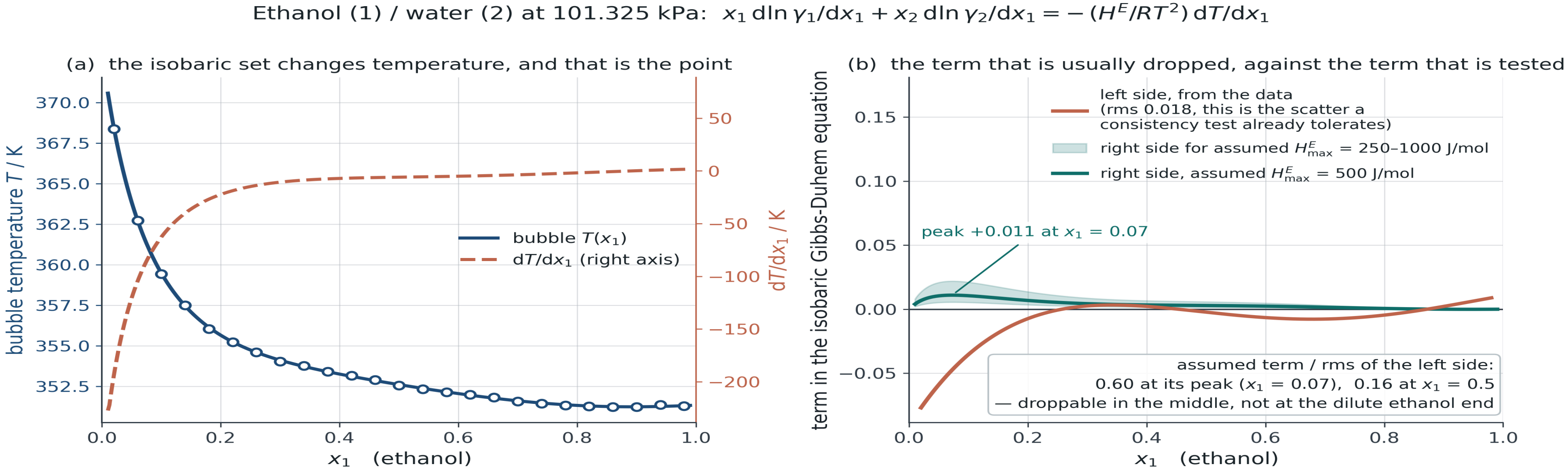
Wisniak L/W

Built on an energy balance rather than on G^E, so it responds to a different combination of the measurements. Worth quoting when a dataset is contested.

Isobaric data: the term that does not vanish

At constant P the temperature varies across the composition range, and the excess enthalpy term returns.

$$x_1 \frac{\mathrm{d}\ln\gamma_1}{\mathrm{d}x_1} + x_2 \frac{\mathrm{d}\ln\gamma_2}{\mathrm{d}x_1} = -\frac{H^E}{RT^2}\,\frac{\mathrm{d}T}{\mathrm{d}x_1}$$

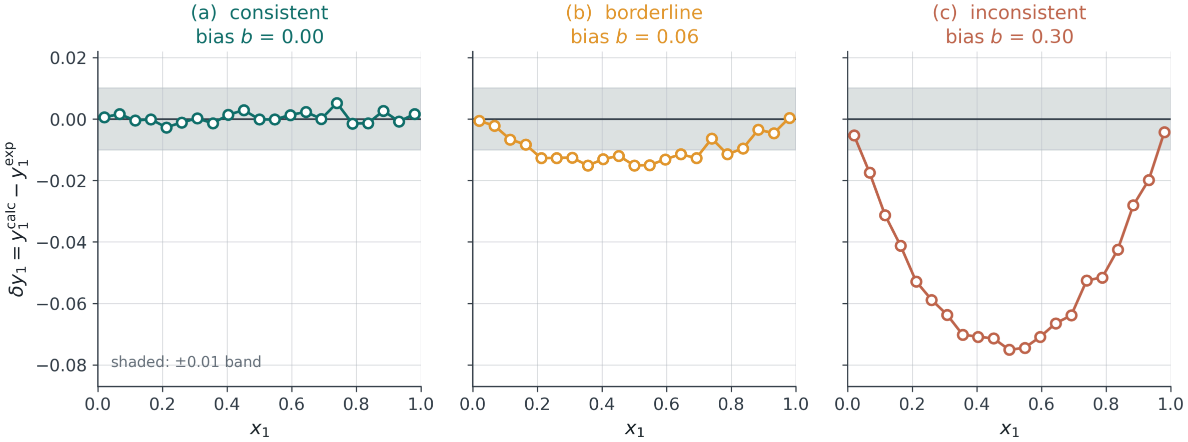


ASSUMED VALUE: $H^E = 4H^E_{max}x_1x_2$ with H^E_{max} = 500 J/mol (band: 250–1000 J/mol). This is an assumed magnitude for an alcohol/water mixture near its boiling range, not a measurement; no calorimetric data are used anywhere in this package. $\mathrm{d}T/\mathrm{d}x_1$ and the left-hand side are computed, from vlekit.regress.bubble_T and vlekit.consistency.gibbs_duhem_residual respectively.

F4.9 · Size of the enthalpy term against the left-hand side, ethanol/water at 101.325 kPa. The H^E magnitude is assumed and labelled as such on the figure.

Three datasets, five tests

2-propanol (1) / 2-butanone (2), isobaric at 101.325 kPa — the same experiment at three bias levels $\delta y_1 = b\,x_1 x_2$



consistency test	threshold	consistent	borderline	inconsistent
area (Redlich-Kister), D	10	0.86	9.66	**47.22**
Herington, $\lvert D - J\rvert$	10	2.05	6.75	**44.31**
point (VNBG), mean $\lvert\delta y_1\rvert$	0.01	0.0014 (index 1)	0.0095 (index 4)	**0.0492 (index 10)**
direct (Van Ness), rms δ	0.05	0.0117 (index 1)	0.0471 (index 2)	**0.2398 (index 10)**
infinite dilution (Kojima)	30	24.0	16.1	**56.0**

Green = pass, red = fail. Every value is what vlekit.consistency.run_all returned for the set plotted above it.
At b = 0.06 every test passes — the area test by 0.3 of its threshold, the point test by 0.0005 — yet the point index is already 4, which is marginal.
Between b = 0.06 and b = 0.08 four of the five verdicts flip together: these are not independent detectors, but one residual read at different thresholds.
Report a dataset with all five, never with the one that happens to pass.

F4.5 · The same experiment at three bias levels, with every statistic tabulated.

Diagnosis, not verdict

A failure is the beginning of an investigation. Each fault has a signature.

Constant offset in δy	Vapour sampling bias — partial condensation, or a GC calibration offset
Residual bowed, one sign throughout	Wrong Antoine constants, or the vapour-phase correction omitted when it mattered
Sign change near the azeotrope	Composition analysis non-linear across the range
Fails at one end only	Impurity concentrating in the volatile or the heavy component
Large $\mathrm{rms}\,\delta P$ as well	The model, not the data. Refit with a more flexible G^E before concluding anything

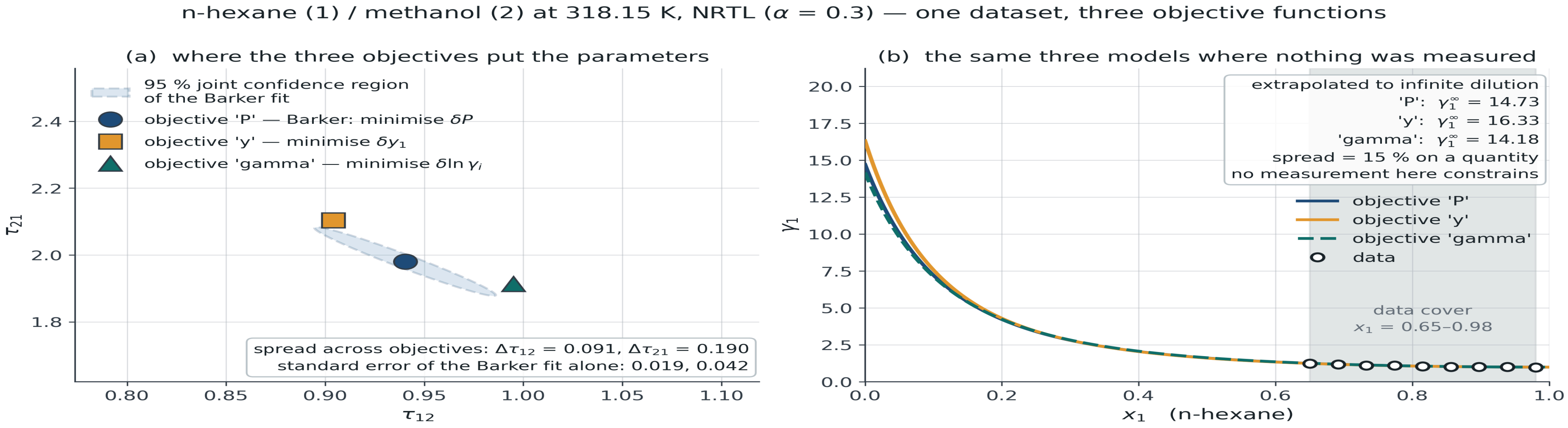
PART V

Building a model from data

Section 4.5 Regression converges to something for almost any input.
The question is never whether a fit exists. It is what the fit is worth.

The objective function is a modelling decision

Same data, same model, four objectives, four answers. This is not numerical noise; it is a choice about which measurement you trust.



The spread is not small

Across objectives the parameters move by several times the standard error that any one of them reports. An uncertainty that ignores the choice of objective is understating the real one.

Minimise what you measured well

Fitting $\ln\gamma$ weights the dilute ends heavily — precisely where the measurement is worst. Fitting y fits the least accurate variable. Fitting P uses the best one.

F4.6 · Parameters and predictions from three objective functions on one dataset.

Barker's method

Fit to P-x alone. Do not fit y at all.

$$P^{\text{calc}}(x_1) = x_1\gamma_1 P_1^{\text{sat}} + x_2\gamma_2 P_2^{\text{sat}}, \qquad \min_{\boldsymbol{\theta}} \sum_k \left(P_k^{\text{calc}} - P_k^{\text{exp}}\right)^2$$

It uses the best data

P and x are the two accurately measured variables. Nothing in the objective depends on the vapour sample.

It makes the tests possible

A y that was never fitted can be compared with a y that was measured. That comparison is the point test and the direct test. Fit y and you destroy your own diagnostic.

Many datasets have no y

Static-cell measurements report P-x only, by design. Barker is not a compromise for those data — it is the intended method.

Does the parameter earn its place?

A model with more parameters cannot fit worse. The residual alone will always choose the larger model.

$$\mathrm{AIC} = n \ln \frac{\mathrm{SSE}}{n} + 2k$$

What AIC answers

Whether the improvement in fit is worth the parameter. A difference below about 2 is conventionally no evidence at all — the data cannot separate those models.

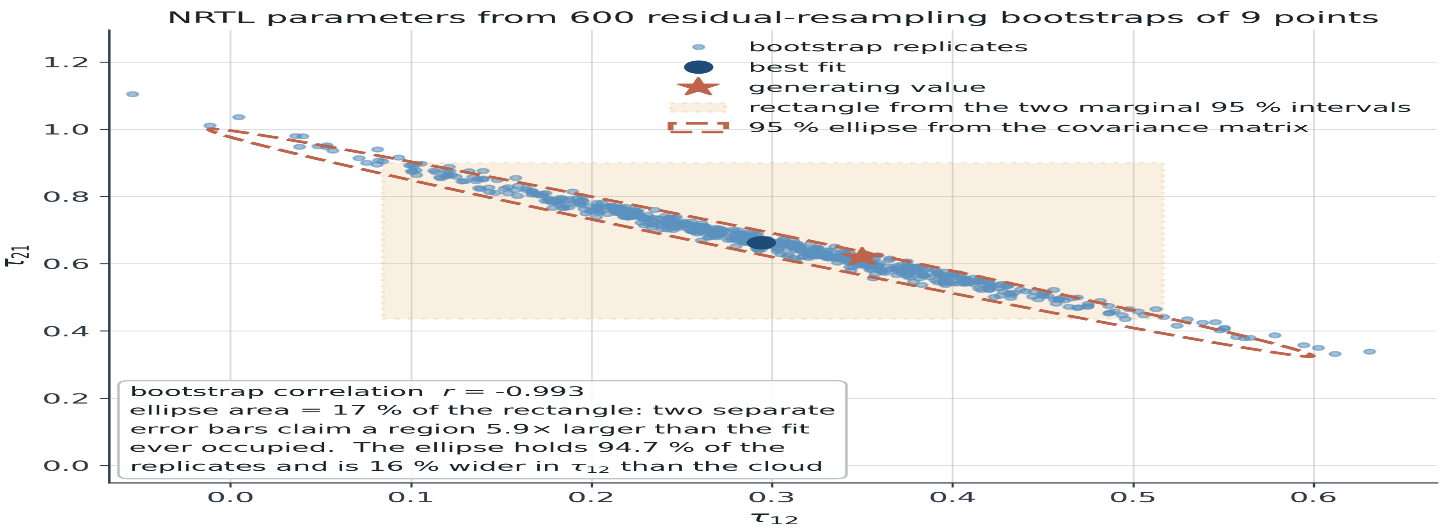
Typical outcome

On twenty-one points at realistic noise, Margules, van Laar, Wilson, NRTL and UNIQUAC land within 2 AIC units of one another. Any claim that one “describes the system better” needs more than that dataset.

AIC compares models fitted the same way. It does ***not*** *compare objective functions: each minimises a different residual vector, so the likelihood being approximated is not the same likelihood.*

Parameter uncertainty is a region, not two numbers

Two standard errors quoted separately describe a rectangle the fit never occupied.



What the cloud shows

A correlation near -1: a long narrow ridge of parameter pairs that fit equally well. The two parameters are not separately identifiable from one isotherm.

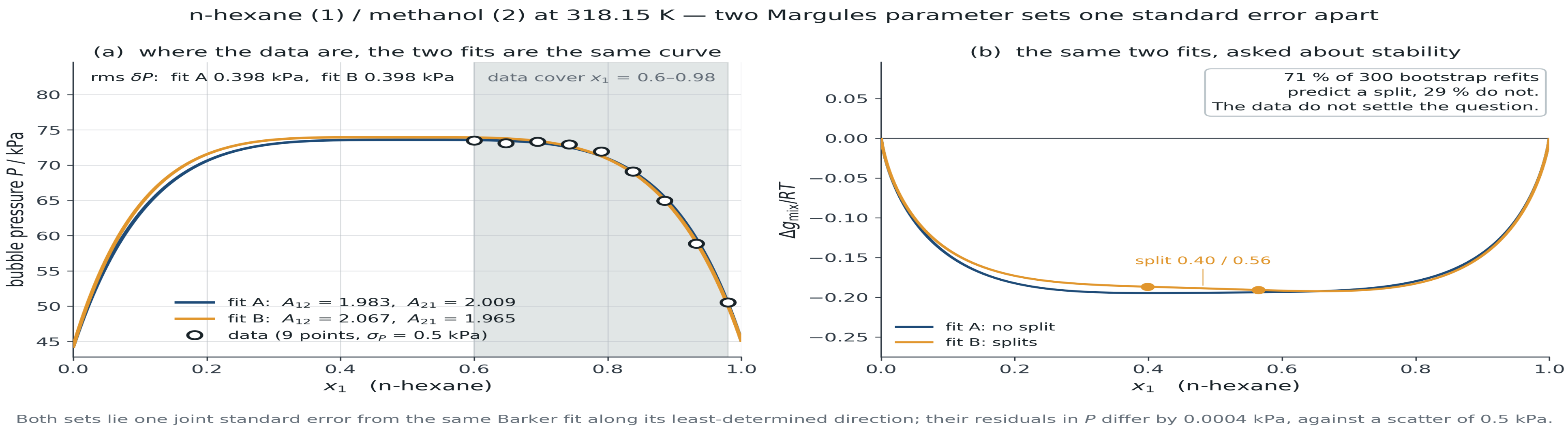
Why bootstrap

The linearised errors assume a quadratic optimum and normal residuals. For two parameters and twenty points, neither holds well. Resampling makes no such assumption and is transparent enough to explain in a report.

F4.7 · Bootstrap replicates with the linearised ellipse over them.

The consequence that actually matters

Two fits the data cannot separate can disagree completely about the region you care about.



Same evidence, different claims

Fitted only where data exist, several models agree to within the scatter. They then report γ^{∞} differing by tens of per cent, and they do not agree on whether the liquid splits into two phases.

Wilson is the clearest case

Wilson is structurally incapable of predicting a miscibility gap for **any** parameters. Fitted to a system that has one, it buys the same fit with an absurd γ^{∞} instead. That number is a symptom of the functional form, not a property of the fluid.

F4.8 · Statistically indistinguishable fits extrapolated beyond the data.

Five models on one real dataset

Ethanol + water, 21 published points. The models agree where the measurements are and disagree about the limit the data barely reach.

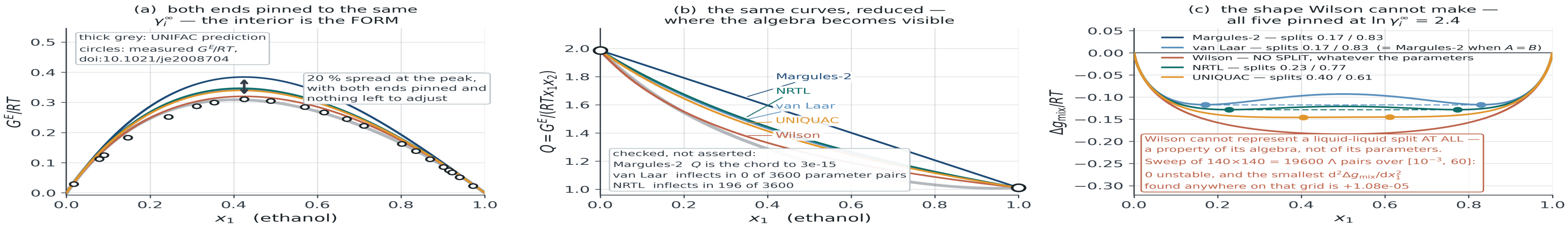
MODEL	Γ_1^∞	Γ_2^∞	RMS ΔY_1	ΔAIC
UNIQUAC	6.00	2.54	0.0042	0
van Laar	5.85	2.51	0.0040	8.7
NRTL	5.77	2.49	0.0039	20.2
Wilson	6.56	2.65	0.0062	49.5
Margules-2	5.19	2.25	0.0068	76.5

The paper reports $u(y_1) = 0.015$. The top three models predict y_1 to 0.004 — ***indistinguishable within the measurement error****, with an AIC ordering driven by differences the data cannot resolve. What separates them is not the fit. It is that γ_1^∞ spans 5.19 to 6.56, a spread of 26 %. Data:* doi:10.1021/je2008704.

Pin the ends, and what is left is the form

Solve each model's two parameters so that every model has *identical* γ_1^∞ and γ_2^∞. Nothing is left to adjust. The curves still differ.

What shape of G^E each model can draw — every model here is a TWO-parameter model with both parameters already spent on the two ends



Nothing on this figure is fitted to data. Each model's two parameters are solved so that both infinite-dilution activity coefficients match a common target exactly; the largest endpoint residual over all five models and both targets is 4.4e-16 in $\ln\gamma^\infty$. Target for (a) and (b): original UNIFAC for ethanol + water at 323.15 K from the published group table in vlekit.unifac — a PREDICTION, not a measurement, plotted with the measured G^E/RT of doi:10.1021/je2008704 for scale. UNIQUAC r, q are UNIFAC group sums, not fitted. NRTL $\alpha = 0.3$, not fitted.

The reduced function $Q = G^E/(RTx_1x_2)$ has $Q(0) = \ln\gamma_1^\infty$ and $Q(1) = \ln\gamma_2^\infty$, so in panel (b) every curve is forced through the same two open circles and only the form is left. Margules-2 is then the chord and has no freedom at all; van Laar is a monotone hyperbola with d^2Q/dx_1^2 of fixed sign. Both statements are identities, and both are checked by sweep rather than asserted.

The cost of choosing a form

With both ends matched to 4×10^{-16} in $\ln\gamma^\infty$, G^E/RT at its peak still spans **20 %** across the five forms — 16 % when the ends are pinned to the NRTL fit instead of to UNIFAC. That residue is the functional form and nothing else.

Same numbers, different function

Margules-2 and van Laar come out with *identical* parameters, 1.7528 and 0.9112: in both, the two parameters **are** $\ln\gamma_1^\infty$ and $\ln\gamma_2^\infty$. Different curves all the same. A parameter value quoted without its model is meaningless.

F4.16 · Nothing on this figure is fitted. Ends pinned to an original-UNIFAC prediction for ethanol + water at 323.15 K, plotted against measured G^E/RT (doi:10.1021/je2008704) for scale.

What each form can draw

Reduce out the ends. $Q = G^E/(RTx_1x_2)$ has $Q(0) = \ln\gamma_1^\infty$ and $Q(1) = \ln\gamma_2^\infty$, so every pinned model passes through the same two points and only the shape is left.

No freedom at all

For Margules-2, Q **is** the straight chord between the two end values — verified to 3×10^{-15} over a 60×60 parameter grid. There is nothing to choose: fix the ends and the whole curve is fixed.

One curvature, fixed sign

Van Laar's Q is a monotone hyperbola: $\mathrm{d}^2Q/\mathrm{d}x_1^2$ changed sign in **0 of 3600** parameter pairs. It can bow away from the chord, but only one way, and it cannot carry an interior feature.

Local composition buys inflection

Wilson inflects in 392 of 3600 pairs, UNIQUAC in 377, NRTL in 196. That is what the extra structure is for — not a better fit at the ends, which any of them can match exactly, but a shape the two-parameter chord cannot make.

This is the honest reason to prefer NRTL or UNIQUAC over Margules for a strongly non-ideal binary, and it has nothing to do with the number of parameters: all five here have exactly two.

Wilson cannot split a liquid

Not “does not usually”. Cannot, for any parameters. This is algebra, not a limitation of fitting.

$$\frac{\mathrm{d}^2}{\mathrm{d}x_1^2}\frac{\Delta g_{\mathrm{mix}}}{RT} = \frac{\Lambda_{12}^2}{x_1(x_1+\Lambda_{12}x_2)^2} + \frac{\Lambda_{21}^2}{x_2(x_2+\Lambda_{21}x_1)^2}$$

Why that settles it

A sum of two squares over positive quantities. Both Λ are ratios of Boltzmann factors and so are positive, hence the second derivative is strictly positive at every composition, hence Δg_{mix} is convex, hence there is no common tangent and no phase split. Ever.

Checked, not asserted

The closed form matches a numerical second derivative to one part in 10^5 over **4000 random** $(\Lambda_{12}, \Lambda_{21}, x_1)$; the smallest value found anywhere is 0, approached only as $\Lambda \to 0$. A 140×140 grid gives 0 unstable pairs of 19600. For contrast, Margules-2 crosses at $A = 2$ exactly and splits above it.

Choosing a form, on shape grounds

Not a capability table. Three questions about your own system, answered before the regression runs.

Is G^E skewed, or symmetric about $x_1 = \frac{1}{2}$?	Symmetric: Margules-2 is enough and its parameters mean something. Strongly skewed: the chord will not do it
Does the reduced Q change curvature?	If it does, only the local-composition forms can follow it. Van Laar and Margules cannot, at any parameter values
Might the system split into two liquid phases, now or under extrapolation?	Wilson is excluded by algebra. NRTL and UNIQUAC can represent a split; whether they should is Module 5
Do you need the parameters at another temperature?	Fit Δu_{ij}, not τ_{ij}, and say what you assumed — three slides on, and F4.9

Two ways to find out whether you have learned the noise

Reporting the residual on the points you fitted is not evidence.

Cross-validation by composition

Train on the middle of the range and predict the dilute ends, then the reverse. Report both. A model that fits its training points three times better than it predicts the held-out ones has learned the scatter.

Splitting on alternate points tests almost nothing — neighbouring points carry almost the same information.

Against a prediction

UNIFAC predicts γ from group contributions with nothing fitted to your data. Your model must beat it — it saw the answer. The question is by how much, and whether the margin survives outside the fitted range.

State the margin in a quantity someone would act on: γ^{∞}, or the azeotrope composition. Not in rms.

Temperature, and the limits of two numbers

Parameters fitted at one temperature, used at another.

$$\tau_{ij} = \exp\left(-\frac{\Delta u_{ij}}{RT}\right) \quad\Longrightarrow\quad \left.\frac{\partial(G^E/RT)}{\partial T}\right|_x = -\frac{H^E}{RT^2}$$

What the extrapolation assumes

Fitting τ directly at one temperature and reusing it assumes G^E/RT is independent of T — that is, $H^E = 0$. Fitting Δu instead assumes Δu is constant, which is weaker but still an assumption.

In practice

Twenty or thirty kelvin is usually tolerable for a non-associating mixture. Across an alcohol/water column from reboiler to condenser it is not, and the error shows up exactly where the column is hardest to control.

The deliverable

A regression report that stops at the parameters is not finished.

1. **Model, parameters and objective.**
   Which G^E form, which two numbers, minimised against what, with which vapour-phase assumption.

2. **Range of validity.**
   The composition, temperature and pressure range of the data, stated as a range and not as a midpoint.

3. **Uncertainty and correlation.**
   A confidence region or a covariance, not two independent error bars.

4. **What was extrapolated.**
   Every quantity you report that lies outside the data — γ^∞, an azeotrope, a predicted phase split — named as an extrapolation.

PART VI

Using the model

Section 4.6 Bubble, dew, flash — and the question none of them asks.

Four calculations, one equation

$y_i P = x_i \gamma_i P_i^{\text{sat}}$ throughout. What changes is which two of P, T, x, y you were given.

BUBL P

Given x, T; solve for P, y.

$$P = \sum_i x_i \gamma_i P_i^{\text{sat}}$$

Everything on the right is known. **Explicit.**

DEW P

Given y, T; solve for P, x.

$$\frac{1}{P} = \sum_i \frac{y_i}{\gamma_i P_i^{\text{sat}}}$$

γ_i needs x, which is the answer.

BUBL T

Given x, P; solve for T, y.

$$\sum_i x_i \gamma_i P_i^{\text{sat}}(T) = P$$

P_i^{sat} is not computable yet.

DEW T

Given y, P; solve for T, x.

$$\sum_i \frac{y_i}{\gamma_i P_i^{\text{sat}}(T)} = \frac{1}{P}$$

Both difficulties at once.

The pattern is in the algebra, not in the physics: the bubble forms are sums, the dew forms are sums of reciprocals. Everything that follows — which one needs a loop, which one is slow, which one fails — comes from that difference and from where the unknown sits.

Which of them needs an inner loop, and why

Two independent difficulties, and they compose. Nothing here is about the solver; it is about which unknown appears inside which function.

Difficulty 1 — γ_i needs the liquid

The activity coefficient is a function of x. In the two DEW problems x is what you are solving for, so γ_i must be re-evaluated every step: successive substitution on x, starting from the ideal answer $x = y$.

Difficulty 2 — P_i^{sat} needs the temperature

Antoine is evaluated at the unknown T in the two T problems, so an outer root-find on T wraps whatever is inside. **DEW T has both**, and the inner loop runs to convergence at every outer trial temperature.

Which is why the cost is multiplicative rather than additive, and why a loose inner tolerance corrupts the outer root-find instead of merely slowing it: the outer function stops being a smooth function of T.

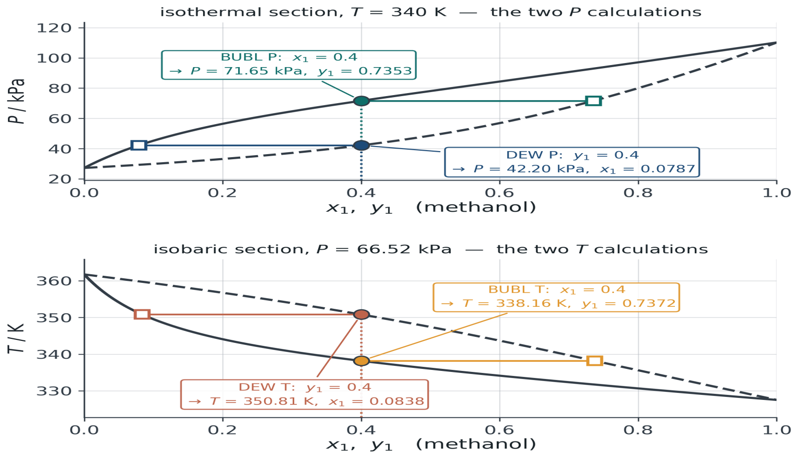
The four, worked

Methanol + water, NRTL fitted by Barker to 16 published points, ideal vapour, composition specified as 0.40 in each case.

BUBL P, DEW P, BUBL T, DEW T: the same equation, specified four ways

one equation, four questions — worked at x_1 or y_1 = 0.4 (methanol)

calculation	fixed	solved for	the equation that closes it	worked answer	what has to iterate, and why
BUBL P	$T,\ x_i$	$P,\ y_i$	$P = \sum_i x_i \gamma_i P_i^{sat}$	P = 71.646 kPa y_1 = 0.7353	explicit — no loop 0 iterations
DEW P	$T,\ y_i$	$P,\ x_i$	$1/P = \sum_i y_i/(\gamma_i P_i^{sat})$	P = 42.197 kPa x_1 = 0.0787	loop on x: γ_i needs the answer 15 composition steps
BUBL T	$P,\ x_i$	$T,\ y_i$	$\sum_i x_i \gamma_i P_i^{sat}(T) = P$	T = 338.163 K y_1 = 0.7372	loop on T: P_i^{sat} not yet computable 9 temperature steps
DEW T	$P,\ y_i$	$T,\ x_i$	$\sum_i y_i/(\gamma_i P_i^{sat}(T)) = 1/P$	T = 350.814 K x_1 = 0.0838	loop on T with the x loop inside it 13 outer × 75 inner steps



methanol (1) + water (2), doi:10.1021/je300613v — NRTL (α = 0.3, τ_{12} = -0.4521, τ_{21} = +1.3988) fitted to the 16 published points by Barker's method, rms δT = 0.163 K. Ideal vapour (Φ_i = 1) throughout; both sections lie inside the 327–362 K range the data cover.
Iteration counts are counted by the solvers in this script, which are written out in full; every converged answer is checked against vlekit.regress / vlekit.flash and agrees to better than 1e-10 in P or T.

The four answers

BUBL P 71.646 kPa, no iteration at all. DEW P 42.197 kPa in 15 composition steps. BUBL T 338.163 K in 9 temperature steps. DEW T 350.814 K in **13 outer × 75 inner**.

The ratio is the lesson

Nothing was made harder: same system, same model, same tolerance. Moving the specification from the liquid to the vapour and from P to T costs about two orders of magnitude in work, and that cost is where a flowsheet fails.

F4.14 · Iteration counts counted by the solvers, not estimated; every answer checked against `vlekit` to better than 10^{-10}. Data: doi:10.1021/je300613v.

The cheapest check you can run

For one composition at one condition, the four answers must be ordered. If they are not, you have a sign error.

The ordering, and why it must hold

At the same T and specified composition, BUBL $P >$ DEW P: 71.646 against 42.197 kPa. At the same P, BUBL $T <$ DEW T: 338.163 against 350.814 K. The bubble point is the first bubble of vapour, the dew point the last drop of liquid, and the two-phase region lies between them.

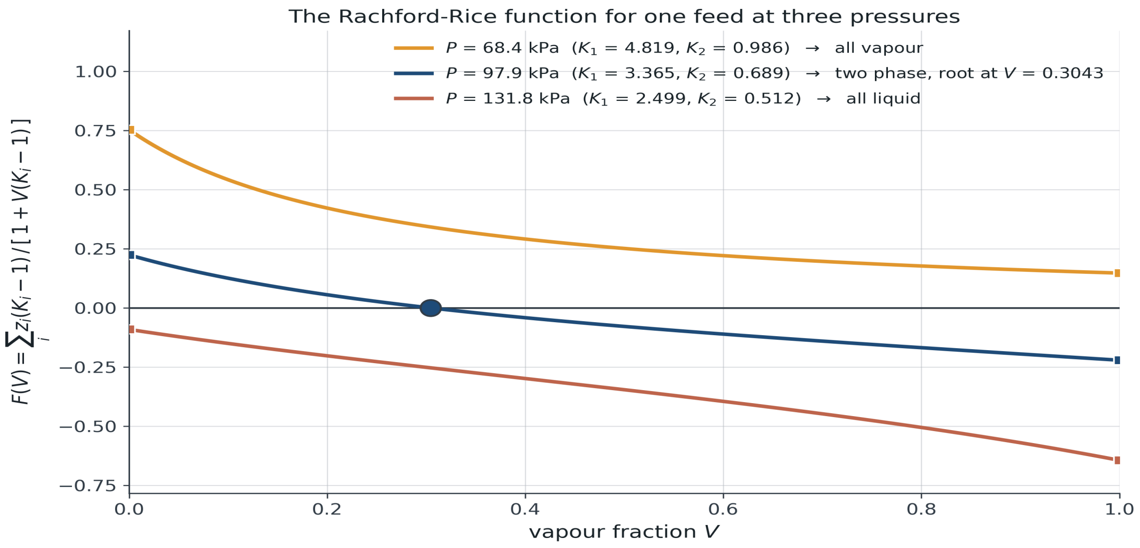
It held on all four systems

Benzene + toluene, methanol + water, ethanol + water and chloroform + 2-butanone: sixteen numbers, ordered correctly in every row, with the from-scratch code and `vlekit` agreeing to 10^{-10}. Two failure modes are caught by one comparison that costs nothing.

Rachford-Rice

Monotonic in the vapour fraction, and therefore safe. Show the plot before the algebra.

$$F(V) = \sum_i \frac{z_i(K_i - 1)}{1 + V(K_i - 1)} = 0, \qquad K_i = \frac{y_i}{x_i} = \frac{\gamma_i P_i^{\mathrm{sat}}}{P}$$



Ethanol (1) / water (2), Margules-2 with A_{12} = 1.6, A_{21} = 0.9 (illustrative, as in F4.2). Feed z_1 = 0.2 at T = 360.00 K; bubble 119.83 kPa, dew 75.96 kPa.
Largest dF/dV over all three curves = -0.126: F is strictly decreasing, so there is at most one root and the signs of $F(0)$ and $F(1)$ settle the phase state before any iteration begins.

F4.10 · One feed at three pressures. The signs of $F(0)$ and $F(1)$ settle the phase state before any iteration begins.

The question the flash never asks

Rachford-Rice was told there are two phases, and it will find two phases.

What is missing

Nothing in the equilibrium equations checks **how many** phases there should be. A fitted model that predicts a liquid-liquid split will still return a single perfectly reasonable-looking liquid composition inside the gap.

What answers it

The Gibbs energy of mixing and a tangent construction — not the equilibrium equations. A liquid splits when Δg_{mix} dips below its own chord. That is Module 5, and it is why the capstone spans both modules.

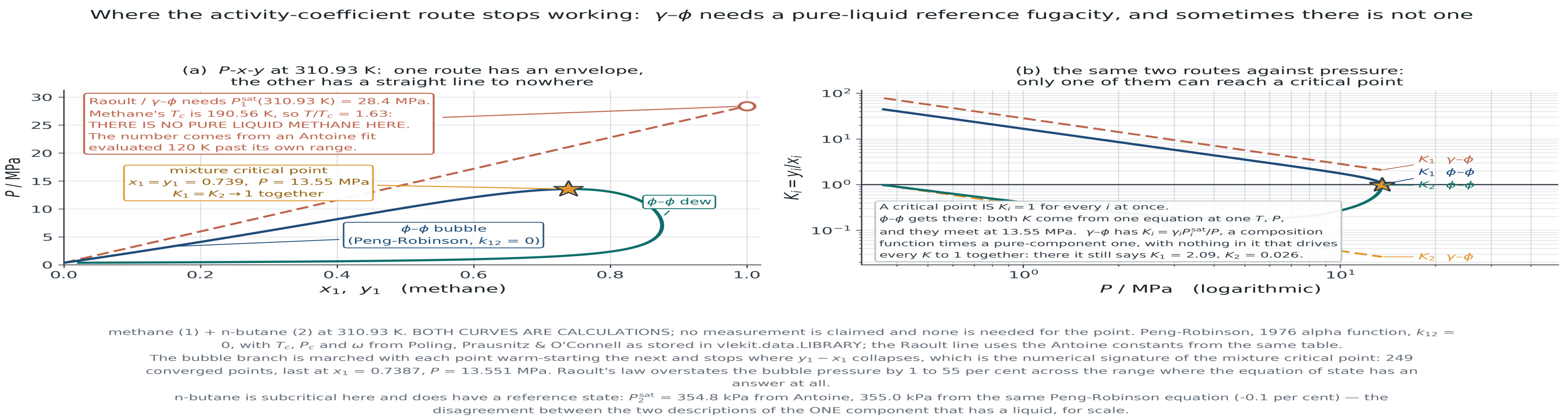
PART VII

The link to equations of state

Section 4.7 The activity route runs out. The equation of state does not describe polar liquids. A mixing rule built from G^E is the bridge — and this section reports what it is actually worth.

Where gamma-phi stops working

Methane + n-butane at 310.93 K. Methane sits at $T/T_c = 1.63$: no pure liquid, so no reference fugacity to write γ against.



What the two routes say at the critical point

The mixture critical point is at $x_1 = y_1 = 0.7387$, 13.551 MPa. Phi-phi gives $K_1 = 1.010$, $K_2 = 0.972$ — going to one together, which is what a critical point *is*. Gamma-phi says 2.09 and 0.026, and cannot make them meet.

The fiction underneath

$P_1^{\text{sat}} = 28.4$ MPa at 310.93 K comes from an Antoine correlation evaluated 120 K past its 91–190 K range: a number, not a vapour pressure.

F4.17 · Both curves are calculations: Peng-Robinson, 1976 α, $k_{12} = 0$.

What the classical mixing rule is secretly assuming

Run the argument backwards. With a quadratic and b linear, the cubic already has an excess Gibbs energy — and you can write it down.

$$\frac{G^E_{\text{cl}}}{RT} = \frac{C}{RT}\left[\frac{a_{\text{cl}}(T,x)}{b_{\text{cl}}(x)} - \sum_i x_i \frac{a_i}{b_i}\right] \xrightarrow{k_{12}=0} \frac{K\, x_1 x_2}{x_1 b_1 + x_2 b_2}$$

It is a van Laar with no freedom

With $k_{12} = 0$ the bracket factorises into a perfect square, so G^E_{cl} is exactly a van Laar function with $A = K/b_2$, $B = K/b_1$ — both constants fixed by the pure-component a_i and b_i. Nothing is adjustable, and $K > 0$, so the classical rule at $k_{12} = 0$ **can only produce positive deviations**.

On a real Type 4 binary

Chloroform + 2-butanone at 303.15 K needs $G^E/RT = -0.278$ at equimolar. The classical rule at $k_{12} = 0$ offers $+0.024$ — the wrong sign. One fitted $k_{12} = -0.072$ buys -0.306, which is why it works here; but the *shape* stays a single skewed lobe with the skew set by $b_1 : b_2 = 63.4 : 82.5$ cm³/mol.

So the classical rule is not an alternative to the G^E route. It ***is*** *the G^E route, with one particular and rather rigid choice of activity model.*

The bridge: Huron-Vidal

Stop guessing $a(x)$. **Require** the equation of state to reproduce a chosen G^E — which can only be done at one reference pressure, because the equation of state's G^E depends on P and the activity model's does not.

$$\frac{a_{\text{mix}}}{b_{\text{mix}}} = \sum_i x_i \frac{a_i}{b_i} + \frac{G^E_\infty}{C}, \qquad b_{\text{mix}} = \sum_i x_i b_i, \qquad C = \frac{\ln(\sqrt{2}-1)}{\sqrt{2}} = -0.6232$$

Why infinite pressure

There $v \to b$, the excess volume vanishes and the expression closes in a and b alone. MHV1 and MHV2 match at zero pressure instead — messier algebra, one large reward, on the last slide of this section.

Watch the sign

C is **negative**, so a positive G^E *lowers* a/b — the direction a positive k_{ij} moves it, which is the check that the convention is right. Some references define C with the opposite sign and then subtract. Both are in print; only one works with the value above. SRK gives $C = -\ln 2$.

What it costs

The reference state is not the real liquid, so **published low-pressure NRTL parameters cannot be reused**, and they do not transfer between SRK and PR because C differs. Unit tests: the pure limits, and G^E from the equation of state converging to the activity model as $P \to \infty$.

Wong-Sandler, and the constraint it exists for

Every cubic has a low-density expansion, and statistical mechanics fixes the composition dependence of its leading term exactly.

$$Z = 1 + \frac{B(T,x)}{v} + \dots, \qquad B(T,x) = b(x) - \frac{a(T,x)}{RT} = \sum_i \sum_j x_i x_j B_{ij}$$

Huron-Vidal violates it

It keeps b linear and puts G^E into a/b, so B inherits the composition dependence of the activity model, which is not a quadratic. Measured on the excess second virial: **2.20 %** for chloroform + 2-butanone, **22.9 %** for ethanol + water — ten times worse, and the violation scales with G^E.

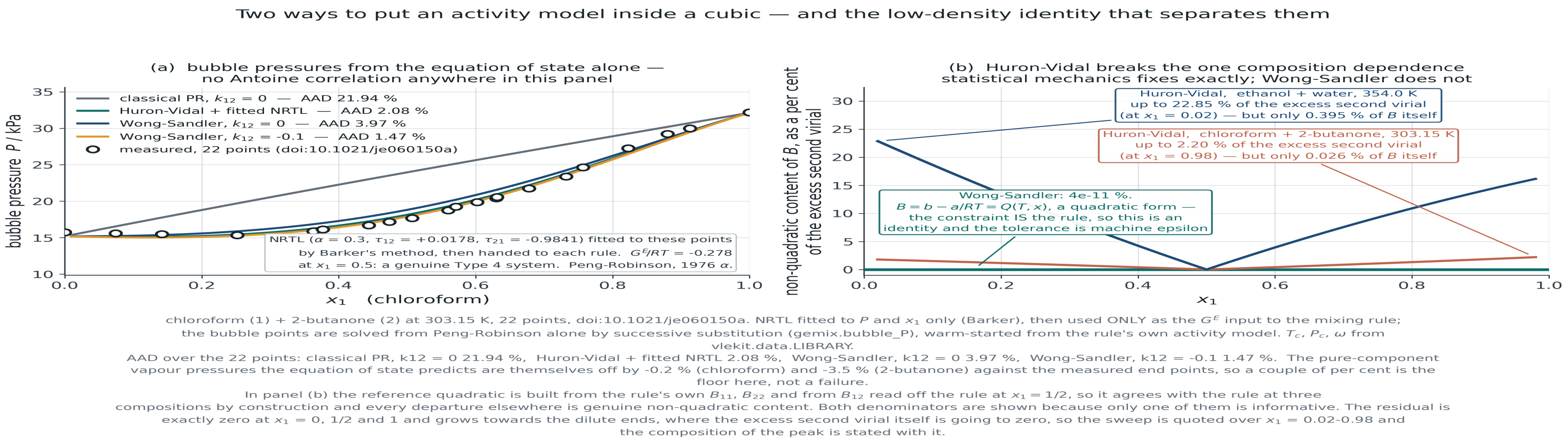
Wong-Sandler imposes it

The quadratic form becomes a second equation, solved together with the G^E match for a **and** b, so b_{mix} stops being linear. Its violation is 4×10^{-11} %: machine precision, because for that rule it is an identity and not an approximation. Cost: one extra binary parameter inside B_{ij}.

The error Huron-Vidal makes is an error in a limit where the cubic ought to be exact. That is the practical form of the complaint that its parameters do not travel in temperature — and it is an argument about a region nobody measured.

What it is actually worth, on real data

Chloroform + 2-butanone at 303.15 K: bubble pressures from the cubic alone.



A factor of ten, for no fitted parameter

Classical Peng-Robinson at $k_{12} = 0$: **21.94 %** AAD in bubble pressure. The same cubic with Huron-Vidal and the Section 4.5 NRTL, which never saw the equation of state: **2.08 %**. Wong-Sandler 3.97 % at $k_{12} = 0$, 1.47 % at $k_{12} = -0.1$.

And the number that loses

Give the classical rule its one parameter, fitted to these pressures: $k_{12} = -0.072$, AAD **1.34 %**. **It beats Huron-Vidal here** — not the result the usual telling leads you to expect.

F4.18 · Peng-Robinson alone; its own pure vapour pressures are -0.2 % and -3.5 % out, so a per cent or two is the floor. doi:10.1021/je060150a

Reading that result honestly

The case for the G^E rules is not that they fit better on one binary. Two things in the table are worth more than the AAD column.

It fits comparably without being fitted here

The Huron-Vidal 2.08 % used NRTL parameters regressed in Section 4.5, which never saw the equation of state and never saw these pressures through it. The classical 1.34 % was regressed against **these very pressures**. Those two numbers are not the same kind of number.

And this binary is an easy case

$G^E/RT = -0.28$ at equimolar: moderately non-ideal, well inside what a single k_{12} can absorb. The claim for the G^E route is that it does not fall apart where the classical rule does — strongly associating mixtures, and extrapolation in temperature, which is where the second-virial violation bites.

ROUTE	FITTED TO THESE PRESSURES?	AAD IN P
Classical PR, $k_{12} = 0$	no	21.94 %
Huron-Vidal + the Section 4.5 NRTL	no	2.08 %
Wong-Sandler, $k_{12} = 0$	no	3.97 %
Wong-Sandler, $k_{12} = -0.1$	one number	1.47 %
Classical PR, $k_{12} = -0.072$	**yes**	**1.34 %**

The last two claims are about conditions this dataset does not contain. Say so, and say what would test them: the same binary at a second temperature, with the classical k_{12} held at the value fitted here.

What the property-method names mean

The dropdown in a process simulator, decoded. The reference state is the thing to read off the name.

PROPERTY METHOD	IS	PARAMETERS
PRWS, RKSWS	Wong-Sandler on PR or SRK	Refit; not the published low-pressure set
PRMHV2, RKSMHV2	Modified Huron-Vidal, second order	Zero-pressure reference — published UNIFAC/UNIQUAC parameters **can** be reused
PSRK	Predictive SRK: MHV1 with UNIFAC	Predictive; no binary VLE data needed
PR, SRK with k_{ij}	The classical rule	One number per pair, fitted, with a range of validity like any other

Choosing a row is choosing a reference state and a parameter obligation, not a level of sophistication. Appendix slides A.2 and A.3 carry the algebra behind the first two rows; A.4 is this table with the derivation references.

What you must be able to do

The module in eight statements.

1. **Choose a description and defend the assumption.**
Raoult, modified Raoult, gamma-phi or phi-phi — and say which assumption you are relying on.

2. **Predict the solution type before fitting.**
Two γ^{∞} and a vapour-pressure ratio decide whether an azeotrope exists, and whether it moves with pressure.

3. **Turn T, P, x, y into γ with its uncertainty.**
Vapour-phase assumption stated, P^{sat} checked against the paper's own end points.

4. **Test a dataset and diagnose the failure.**
All five tests, and a signature-based diagnosis rather than a verdict.

5. **Fit, and say what the fit is worth.**
Objective chosen deliberately, confidence region reported, extrapolations named — and compared against UNIFAC.

6. **Choose the functional form on shape grounds.**
With both ends pinned, 20 % of G^E is still the form; Wilson cannot split a liquid at all.

7. **Run all four bubble and dew calculations.**
Know which need an inner loop, and check the ordering before trusting the answer.

8. **Know where the activity route stops.**
No reference fugacity above T_c; a G^E mixing rule is the bridge, at a stated reference state.

Appendix

EoS-G^E mixing rules
the derivations, in full

The algebra behind Section 4.7 — and the unit tests that settle it.

A.1 · Two worlds that do not meet

Equations of state

Handle high pressure and supercritical components. One model for both phases, no reference vapour pressure required. Fail on polar and hydrogen-bonding mixtures, because the quadratic mixing rule assumes random mixing.

Activity coefficient models

Handle polar and associating liquids, through local-composition terms fitted to data. Cannot cross the critical point, and need a reference fugacity that a supercritical component does not have.

The engineering problem is the mixture that is both — CO_2 with ethanol, water with a light gas — and neither description can be pushed into the other's territory. Section 4.7 builds the bridge; the two slides that follow carry the algebra it quotes.

A.2 · The Huron-Vidal algebra

Where C comes from, in five lines. Section 4.7 quotes the result; this is the limit it is quoted from.

$$\frac{G^{\mathrm{res}}}{RT} = B - \frac{A}{2\sqrt{2}B} \ln \frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B} + Z - 1 - \ln Z$$

$$P \to \infty : \quad Z \to B, \quad \ln \frac{2+\sqrt{2}}{2-\sqrt{2}} = 2\ln(1+\sqrt{2}), \quad \frac{G^{\mathrm{res}}}{RT} \to B + C\,\frac{a}{bRT}$$

Then subtract the pure components

$$\frac{G^E_\infty}{RT} = \frac{P}{RT}\left(b - \textstyle\sum_i x_i b_i\right) + C\left[\frac{a}{bRT} - \textstyle\sum_i x_i \frac{a_i}{b_i RT}\right]$$

The first bracket is PV^E and vanishes **iff b is linear in composition** — exactly the assumption Huron-Vidal makes and exactly the one Wong-Sandler gives up. With it gone, $a/b = \sum_i x_i a_i/b_i + G^E_\infty/C$.

The sign, done slowly

$C = -\ln(1+\sqrt{2})/\sqrt{2} = \ln(\sqrt{2}-1)/\sqrt{2} = -0.6232$ — the same number written two ways, and negative. A positive G^E therefore lowers a/b, which is the direction a positive k_{ij} moves it. Writing the rule as "$-\,G^E/C$" with a negative C turns every positive-deviation system into a negative-deviation one. Settle it with a numerical limit, not with a reference.

Huron, M.; Vidal, J. Fluid Phase Equilib. **3** *(1979) 255-271. SRK gives $C = -\ln 2$; the constant depends on the volume translation of the cubic and on nothing else.*

A.3 · The Wong-Sandler algebra, and the trap

Two conditions, two unknowns. Then the test that makes a correct implementation look broken.

$$Q = \sum_i \sum_j x_i x_j \left(b - \frac{a}{RT}\right)_{ij}, \qquad D = \sum_i x_i \frac{a_i}{b_i RT} + \frac{G^E_\infty}{CRT}$$

$$b_{\text{mix}} = \frac{Q}{1 - D}, \qquad a_{\text{mix}} = RT\,\frac{Q\,D}{1 - D}$$

Why b stops being linear

Q is the imposed quadratic second virial and D carries the G^E match, so $b_{\text{mix}} = Q/(1 - D)$ inherits the composition dependence of both. That is the structural difference from Huron-Vidal, and it is what buys $B(T, x) = \sum_i \sum_j x_i x_j B_{ij}$ as an identity.

The trap

The infinite-pressure derivation matches an excess **Helmholtz** energy. $G^E - A^E = PV^E$; Huron-Vidal has $V^E_\infty = 0$ so the two coincide, but Wong-Sandler's b_{mix} is not linear, so its G^E from the equation of state **diverges** like PV^E while its A^E converges. At $P = 10^{11}$ kPa and $x_1 = 0.4$: $A^E/RT = -0.258836$ against the activity model's -0.258836, while G^E/RT reads -93713.

Wong, D. S. H.; Sandler, S. I. AIChE J. ***38*** *(1992) 671-680. Pure-component reduction ($b_{\text{mix}} = b_i$, $a_{\text{mix}} = a_i$) is the first unit test; the A^E limit above is the second.*

A.4 · Before you trust an implementation

Five checks, each a limit the algebra must reproduce exactly. Every number below came out of a run, not out of a reference.

Pure-component reduction

$b_{\text{mix}} = b_i$ and $a_{\text{mix}} = a_i$ at $x_i = 1$, for both rules. Catches most indexing errors immediately

Huron-Vidal, $P \to \infty$

G^E/RT from the equation of state $= -0.258836$ at $x_1 = 0.4$, against the activity model's -0.258836. Converged by 10^{11} kPa

Wong-Sandler, $P \to \infty$

Test A^E, not G^E: -0.258836 against the same target, while G^E/RT reads -93713 and is *supposed* to

Second virial

Wong-Sandler reproduces $\sum_i \sum_j x_i x_j B_{ij}$ to 4×10^{-11} % of the excess; Huron-Vidal to 2.20 % and 22.9 %

The classical rule as a special case

Feed the equation of state's own G^E back into either rule and the classical result returns. At $k_{12} = 0$ it is a van Laar: $A = 0.0840$, $B = 0.1093$ for chloroform + 2-butanone at 303.15 K

All five are in `vlekit/gemix.py` and its test module. MHV1 and MHV2 (Michelsen, 1990) match at zero pressure instead and are not implemented in the course toolkit — the reason published UNIFAC parameters transfer to PSRK is the reference state, and that argument is on the taught slide.